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TREND-BASED MOCK PAPER

GATE Ecology and Evolution

Mock Paper — 03

Full-Length Examination Simulation

Duration: 3 Hours | Maximum Marks: 100 | Questions: 65

Examination Details

Branch:	Ecology and Evolution
Paper:	Mock 03
Total Questions:	65
Total Marks:	100
Duration:	3 Hours
Question Types:	MCQ, MSQ, NAT
Negative Marking:	Yes (MCQ only)
Difficulty:	Mixed

*Unofficial practice material. Not affiliated with or endorsed by GATE/IIT.
Historical patterns do not guarantee future examination questions.
This is a trend-based practice simulation.*

GENERAL INSTRUCTIONS

1. This question paper contains 65 questions for a total of 100 marks.
2. Duration of examination: 3 Hours.
3. The question types include:
 - MCQ (Single Correct Answer): 1 or 2 marks each
 - MSQ (Multiple Correct Answers): 2 marks each
 - NAT (Numerical Answer Type): 1 or 2 marks each
4. For MCQ: Only one option is correct. Select the single correct answer.
5. For MSQ: One or more options are correct. Select ALL correct options.
6. For NAT: Enter the numerical answer. No negative marking for NAT questions.
7. Negative marking: -1/3 mark for 1-mark MCQ (wrong answer), -2/3 mark for 2-mark MCQ.
No negative marking for MSQ and NAT questions.
8. A virtual calculator is available on the screen. Physical calculators are not permitted.
9. The question paper contains 8 pages including this instructions page.
10. All questions are compulsory.
11. Fill in the required details on the answer sheet before starting.
12. Do not ask for clarifications from the invigilator. Interpret questions as given.

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Section A: General Aptitude

Q1. [1-MCQ] [1 mark]

Antonym of 'indigenous':

- (A) Native
- (B) Foreign
- (C) Local
- (D) Original

Section B: Ecology and Evolution

Q2. [1-NAT] [1 mark]

Standard deviation is:

(Numerical Answer Type — Enter the numerical value)

Q3. [1-MCQ] [1 mark]

The vector (1,2,3) projected onto (2,1,0) is:

- (A) (4/5, 2/5, 0)
- (B) (8/5, 4/5, 0)
- (C) (2, 1, 0)
- (D) (1, 2, 3)

Q4. [1-NAT] [1 mark]

Which of the following is a probability generating function for Poisson(λ)?

(Numerical Answer Type — Enter the numerical value)

Q5. [1-NAT] [1 mark]

A random variable $X \sim \text{Poisson}(\lambda=4)$. $P(X \geq 2)$ is approximately:

(Numerical Answer Type — Enter the numerical value)

Q6. [1-MCQ] [1 mark]

If $f(x) = x^3 - 3x + 2$, number of real roots:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q7. [1-MCQ] [1 mark]

$P(|Z| \leq 1)$ for $Z \sim N(0,1)$ is approximately:

- (A) 0.68
- (B) 0.32
- (C) 0.95
- (D) 0.16

Q8. [1-MCQ] [1 mark]

Gauss-Seidel method converges when the coefficient matrix is:

- (A) Symmetric
- (B) Positive definite
- (C) Diagonally dominant
- (D) Non-singular

Q9. [1-MCQ] [1 mark]

$d/dx[\ln(\sec x) + \ln(\tan x)]$ evaluated at $x = \pi/4$ equals:

- (A) 0
- (B) 2
- (C) 4
- (D) 1

Q10. [1-MCQ] [1 mark]

For a Poisson distribution with mean 4, variance is:

- (A) 4
- (B) 2
- (C) 16
- (D) 1

Q11. [1-MCQ] [1 mark]

Solution of $y'' + 4y = 0$ with $y(0)=1$, $y'(0)=0$, $y(\pi/2)$ equals:

- (A) 1
- (B) 0
- (C) -1
- (D) $\cos(\pi)$

Q12. [2-MCQ] [2 marks]

Cauchy-Schwarz inequality states $|\langle u, v \rangle| \leq$:

- (A) $\|u\| * \|v\|$
 - (B) $\|u\| + \|v\|$
 - (C) $\|u\| / \|v\|$
 - (D) $\sqrt{\|u\| * \|v\|}$
-

Q13. [2-NAT] [2 marks]

Fourier series of $f(x)=x$ on $(-\pi,\pi)$ has only:

(Numerical Answer Type — Enter the numerical value)

Q14. [2-NAT] [2 marks]

If a 3×3 matrix has eigenvalues 1, 2, and 3, then $\text{trace}(A^2)$ equals:

(Numerical Answer Type — Enter the numerical value)

Q15. [2-MCQ] [2 marks]

Laplace transform of $e^{(-3t)}u(t)$ is:

- (A) $1/(s+3)$
 - (B) $1/(s-3)$
 - (C) $1/s$
 - (D) e^{-3}/s
-

Q16. [2-MCQ] [2 marks]

$\lim_{n \rightarrow \infty} (1 + 1/n)^{2n}$ equals:

- (A) e
 - (B) e^2
 - (C) 1
 - (D) infinity
-

Q17. [2-MCQ] [2 marks]

Consider $f(x) = |x|$ on $[-1,1]$. This function is:

- (A) Differentiable everywhere
 - (B) Continuous but not differentiable at 0
 - (C) Neither continuous nor differentiable
 - (D) Differentiable at 0 only
-

Q18. [2-MCQ] [2 marks]

Rolle's theorem requires $f(x)$ to be:

- (A) Continuous only
 - (B) Differentiable only
 - (C) Continuous on $[a,b]$ and differentiable on (a,b)
 - (D) Both continuous and differentiable everywhere
-

Q19. [2-MCQ] [2 marks]

For matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $A^{50} \cdot \begin{bmatrix} 1 & 0 \end{bmatrix}^T$ equals:

- (A) $\begin{bmatrix} 1 & 0 \end{bmatrix}^T$
 - (B) $\begin{bmatrix} 2 & 1 \end{bmatrix}^T$
 - (C) $[(3+2\sqrt{2})^{50}, \dots]$
 - (D) Cannot determine
-

Q20. [2-MCQ] [2 marks]

The area of a circle of radius r is:

- (A) πr^2
- (B) $2\pi r$
- (C) πr
- (D) $2\pi r^2$

Q21. [2-MCQ] [2 marks]

If $f(z) = z^2 + 3z + 2$ is analytic, then $\operatorname{Re}(f(i)) =$

- (A) -1
- (B) -2
- (C) 1
- (D) 2

Q22. [2-NAT] [2 marks]

The differential equation $y \, dx - x \, dy = 0$ has solution:

(Numerical Answer Type — Enter the numerical value)

Q23. [2-MCQ] [2 marks]

Number of spanning trees in complete graph K_4 :

- (A) 4
- (B) 8
- (C) 16
- (D) 12

Q24. [2-NAT] [2 marks]

The rank of the matrix $\begin{bmatrix} 1, 2, 3 \\ 2, 4, 6 \\ 3, 6, 9 \end{bmatrix}$ is:

(Numerical Answer Type — Enter the numerical value)

Q25. [2-NAT] [2 marks]

$P(A \cup B) = 0.6$, $P(A \cap B) = 0.2$, $P(B) = 0.4$. Then $P(A|B)$ equals:

(Numerical Answer Type — Enter the numerical value)

Q26. [2-NAT] [2 marks]

The Laplace transform of $t \cdot e^{-2t}$ is:

(Numerical Answer Type — Enter the numerical value)

Q27. [2-MCQ] [2 marks]

For the ODE $\frac{dy}{dx} = \frac{y}{x} + \sqrt{\frac{y^2}{x^2}}$, with $y(1)=0$, the solution is:

- (A) $y = 0$
- (B) $y = x \ln(x)$
- (C) $y = x \sin(\ln x)$
- (D) $y = x^2$

Q28. [2-MCQ] [2 marks]

The area of a circle of radius r is:

- (A) πr^2
- (B) $2\pi r$
- (C) πr
- (D) $2\pi r^2$

Q29. [2-NAT] [2 marks]

Laplace transform of $e^{-3t} \cdot u(t)$ is:

(Numerical Answer Type — Enter the numerical value)

Q30. [2-MCQ] [2 marks]

Cauchy-Schwarz inequality states $|\langle u, v \rangle| \leq \sqrt{\langle u, u \rangle \langle v, v \rangle}$:

- (A) $\|u\| * \|v\|$
- (B) $\|u\| + \|v\|$
- (C) $\|u\| / \|v\|$
- (D) $\sqrt{\|u\| * \|v\|}$

Q31. [2-MCQ] [2 marks]

Consider $f(x) = |x|$ on $[-1,1]$. This function is:

- (A) Differentiable everywhere
- (B) Continuous but not differentiable at 0
- (C) Neither continuous nor differentiable
- (D) Differentiable at 0 only

Q32. [2-MSQ] [2 marks]

The area of a circle of radius r is:

- (A) $\pi * r^2$
- (B) $2 * \pi * r$
- (C) $\pi * r$
- (D) $2 * \pi * r^2$

Q33. [1-MCQ] [1 mark]

Logistic growth has:

- (A) No limit
- (B) Carrying capacity
- (C) Exponential
- (D) None

Q34. [1-MCQ] [1 mark]

Trophic level energy transfer efficiency:

- (A) 10%
- (B) 50%
- (C) 90%
- (D) 1%

Q35. [1-MCQ] [1 mark]

Most abundant greenhouse gas:

- (A) CO₂
- (B) CH₄
- (C) N₂O
- (D) Water vapor

Q36. [1-MCQ] [1 mark]

Logistic growth equation has carrying capacity:

- (A) $dN/dt = rN$
- (B) $dN/dt = rN(1-N/K)$
- (C) $dN/dt = r$
- (D) $dN/dt = K-N$

Q37. [2-MCQ] [2 marks]

Trophic level energy transfer efficiency:

- (A) 10%
- (B) 50%
- (C) 90%

(D) 1%

Q38. [2-MCQ] [2 marks]

Logistic growth equation has carrying capacity:

- (A) $dN/dt = rN$
- (B) $dN/dt = rN(1-N/K)$
- (C) $dN/dt = r$
- (D) $dN/dt = K-N$

Q39. [2-NAT] [2 marks]

Logistic growth equation has carrying capacity:

(Numerical Answer Type — Enter the numerical value)

Q40. [2-MSQ] [2 marks]

Logistic growth equation has carrying capacity:

- (A) $dN/dt = rN$
- (B) $dN/dt = rN(1-N/K)$
- (C) $dN/dt = r$
- (D) $dN/dt = K-N$

Q41. [2-MSQ] [2 marks]

Trophic level energy transfer efficiency:

- (A) 10%
- (B) 50%
- (C) 90%
- (D) 1%

Q42. [1-MCQ] [1 mark]

Hardy-Weinberg needs:

- (A) No mutation
- (B) No migration
- (C) Large pop
- (D) All

Q43. [1-NAT] [1 mark]

AaBb x aabb. Genotypic ratio:

(Numerical Answer Type — Enter the numerical value)

Q44. [1-MCQ] [1 mark]

mRNA processing:

- (A) Transcription only
- (B) Capping, splicing, poly-A
- (C) Translation only
- (D) Replication

Q45. [1-NAT] [1 mark]

DNA B-form base pairs per turn:

(Numerical Answer Type — Enter the numerical value)

Q46. [2-NAT] [2 marks]

ABO blood group has how many alleles?

(Numerical Answer Type — Enter the numerical value)

Q47. [2-MCQ] [2 marks]

Linked genes show:

- (A) Independent assortment
 - (B) No recombination
 - (C) Less than 50% recombination
 - (D) Always 50%
-

Q48. [2-MCQ] [2 marks]

Linked genes show:

- (A) Independent assortment
 - (B) No recombination
 - (C) Less than 50% recombination
 - (D) Always 50%
-

Q49. [2-MCQ] [2 marks]

Linked genes show:

- (A) Independent assortment
 - (B) No recombination
 - (C) Less than 50% recombination
 - (D) Always 50%
-

Q50. [2-MSQ] [2 marks]

ABO blood group has how many alleles?

- (A) 2
 - (B) 3
 - (C) 4
 - (D) 6
-

Q51. [1-MCQ] [1 mark]

DNA replication in eukaryotes occurs in:

- (A) Cytoplasm
 - (B) Nucleus
 - (C) Mitochondria only
 - (D) G1 phase
-

Q52. [1-MCQ] [1 mark]

Photosynthesis light reactions produce:

- (A) Glucose only
 - (B) ATP and NADPH
 - (C) CO₂
 - (D) O₂ only
-

Q53. [1-MCQ] [1 mark]

Mendelian dihybrid cross (both heterozygous) ratio:

- (A) 3:1
- (B) 9:3:3:1
- (C) 1:2:1

(D) 1:1:1:1

Q54. [1-MCQ] [1 mark]

Competitive enzyme inhibition: K_m

- (A) Increases
- (B) Decreases
- (C) Unchanged
- (D) Zero

Q55. [1-MCQ] [1 mark]

Central dogma: reverse transcription is:

- (A) RNA->DNA
- (B) DNA->RNA
- (C) Protein->DNA
- (D) RNA->Protein

Q56. [2-NAT] [2 marks]

Competitive enzyme inhibition: K_m

(Numerical Answer Type — Enter the numerical value)

Q57. [2-MCQ] [2 marks]

Photosynthesis light reactions produce:

- (A) Glucose only
- (B) ATP and NADPH
- (C) CO₂
- (D) O₂ only

Q58. [2-NAT] [2 marks]

DNA replication in eukaryotes occurs in:

(Numerical Answer Type — Enter the numerical value)

Q59. [2-MCQ] [2 marks]

Competitive enzyme inhibition: K_m

- (A) Increases
- (B) Decreases
- (C) Unchanged
- (D) Zero

Q60. [1-NAT] [1 mark]

Darwin finches from:

(Numerical Answer Type — Enter the numerical value)

Q61. [1-MCQ] [1 mark]

Biodiversity hotspot criteria:

- (A) Endemism
- (B) Deforestation
- (C) Both
- (D) None

Q62. [1-NAT] [1 mark]

Mitochondria function:

(Numerical Answer Type — Enter the numerical value)

Q63. [1-MCQ] [1 mark]

Photosynthesis in:

- (A) Mitochondria
 - (B) Chloroplast
 - (C) Nucleus
 - (D) Ribosome
-

Q64. [1-NAT] [1 mark]

Binomial nomenclature by:

(Numerical Answer Type — Enter the numerical value)

Q65. [1-MCQ] [1 mark]

Monarch butterfly migrates from:

- (A) Canada to Mexico
 - (B) India
 - (C) Africa
 - (D) Europe
-

ANSWER KEY

Q. No.	Type	Answer	Marks
1	1MCQ	B	1
2	1NAT	B	1
3	1MCQ	B	1
4	1NAT	A	1
5	1MCQ	D	1
6	1MCQ	C	1
7	1NAT	B	1
8	1MCQ	B	1
9	1NAT	A	1
10	1MCQ	A	1
11	1MCQ	A	1
12	1MCQ	B	1
13	1NAT	A	1

14	1MCQ	D	1
15	1NAT	A	1
16	1MCQ	B	1
17	1MCQ	B	1
18	1MCQ	B	1
19	1MCQ	B	1
20	1NAT	A	1
21	1MCQ	B	1
22	1MCQ	A	1
23	1MCQ	A	1
24	1MCQ	A	1
25	1MCQ	C	1
26	1MCQ	C	1
27	1MCQ	A	1
28	1MCQ	B	1

29	1NAT	B	1
30	1MCQ	C	1
31	2MCQ	A	2
32	2NAT	B	2
33	2NAT	A	2
34	2NAT	A	2
35	2MCQ	A	2
36	2MCQ	B	2
37	2MCQ	C	2
38	2MCQ	B	2
39	2MCQ	C	2
40	2MCQ	A	2
41	2MCQ	A	2
42	2MCQ	A	2
43	2NAT	B	2

44	2MCQ	C	2
45	2NAT	A	2
46	2NAT	B	2
47	2NAT	A	2
48	2MCQ	A	2
49	2MCQ	A	2
50	2MCQ	A	2
51	2MCQ	B	2
52	2NAT	A	2
53	2NAT	A	2
54	2MCQ	C	2
55	2MCQ	A	2
56	2MCQ	B	2
57	2MCQ	B	2
58	2NAT	B	2

59	2MCQ	C	2
60	2MSQ	B	2
61	2NAT	B	2
62	2MSQ	A	2
63	2MSQ	A	2
64	2MSQ	B	2
65	2MCQ	A	2

DETAILED SOLUTIONS

Q1.

Answer: B

Solution:

Correct Answer: B

Subject: General Aptitude | Topic: Verbal | Difficulty: moderate

Q2.

Answer: B

Solution:

Correct Answer: B

Subject: Engineering Mathematics | Topic: Statistics | Difficulty: moderate

Q3.

Answer: B

Solution:

Correct Answer: B

Subject: Ecology | Topic: Population | Difficulty: moderate

Q4.

Answer: A

Solution:

Correct Answer: A

Subject: Evolution | Topic: Natural Selection | Difficulty: moderate

Q5.

Answer: D

Solution:

Correct Answer: D

Subject: Genetics | Topic: Population Genetics | Difficulty: moderate

Q6.

Answer: C

Solution:

Correct Answer: C

Subject: Environmental Science | Topic: Biodiversity | Difficulty: easy

Q7.

Answer: B

Solution:

Correct Answer: B

Subject: Cell Biology | Topic: Mitochondria | Difficulty: moderate

Q8.

Answer: B

Solution:

Correct Answer: B

Subject: Plant Physiology | Topic: Photosynthesis | Difficulty: easy

Q9.

Answer: A

Solution:

Correct Answer: A

Subject: Zoology | Topic: Taxonomy | Difficulty: moderate

Q10.

Answer: A

Solution:

Correct Answer: A

Subject: Behavioral Ecology | Topic: Migration | Difficulty: easy

Q11.

Answer: A

Solution:

10% law: ~10% energy transferred between trophic levels (Lindeman).

Subject: Ecology | Topic: Ecology | Difficulty: moderate

Q12.

Answer: B

Solution:

$\text{proj} = ((v \cdot w)/(w \cdot w)) \cdot w = (4/5) \cdot (2, 1, 0) = (8/5, 4/5, 0)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q13.

Answer: A

Solution:

AaBb x aabb: offspring AaBb, Aabb, aaBb, aabb in 1:1:1:1.

Subject: Genetics | Topic: Genetics | Difficulty: easy

Q14.

Answer: D

Solution:

Water vapor: most abundant GHG by concentration and radiative forcing.

Subject: Ecology | Topic: Ecology | Difficulty: easy

Q15.

Answer: A

Solution:

PGF of Poisson(λ): $G(z) = E[z^X] = \exp(\lambda(z-1))$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q16.

Answer: B

Solution:

DNA replication: S phase in nucleus. Mitochondrial DNA replicates separately.

Subject: Biology | Topic: Biology | Difficulty: moderate

Q17.

Answer: B

Solution:

Light reactions: ATP + NADPH for Calvin cycle. O₂ is byproduct.

Subject: Biology | Topic: Biology | Difficulty: moderate

Q18.

Answer: B

Solution:

Logistic: $dN/dt = rN(1-N/K)$. K is carrying capacity.

Subject: Ecology | Topic: Ecology | Difficulty: easy

Q19.

Answer: B

Solution:

AaBb x AaBb: phenotypic ratio 9:3:3:1.

Subject: Biology | Topic: Biology | Difficulty: moderate

Q20.

Answer: A

Solution:

$P(X \geq 2) = 1 - e^{-4}(1+4) = 1 - 5e^{-4} \approx 0.908$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q21.

Answer: B

Solution:

$f(0)=2, f(1)=0, f(-1)=4, f(2)=4, f(-2)=-8+6+2=0$. Roots at $x=1$ (double?) and $x=-2$. $f(x)=(x-1)^2(x+2)$. Two distinct real roots.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q22.

Answer: A

Solution:

Competitive: V_{max} unchanged, K_m increases (more substrate needed).

Subject: Biology | Topic: Biology | Difficulty: difficult

Q23.

Answer: A

Solution:

Standard normal: $P(-1 < Z < 1)$ approx 0.68.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q24.

Answer: A

Solution:

Reverse transcription: RNA $\rightarrow$ DNA (retroviruses like HIV).

Subject: Biology | Topic: Biology | Difficulty: moderate

Q25.

Answer: C

Solution:

Gauss-Seidel converges if matrix is diagonally dominant. Jacobi needs symmetric positive definite.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q26.

Answer: C

Solution:

$\frac{d}{dx}[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1 + 2 = 3$. Hmm. Let me recalc: $\frac{d}{dx} \ln(\sec x) = \tan x$. $\frac{d}{dx} \ln(\tan x) = \sec^2 x / \tan x = 1/(\sin x \cos x) = 2\csc(2x)$. At $\pi/4$: $1 + 2 = 3$. Not in options. The derivative of $\ln(\sec x * \tan x) = \frac{d}{dx}[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1+2=3$. Hmm.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q27.

Answer: A

Solution:

Poisson: mean = λ = variance. Both = 4.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q28.

Answer: B

Solution:

Pre-mRNA: 5 cap, intron splicing, 3 poly-A tail.

Subject: Genetics | Topic: Genetics | Difficulty: moderate

Q29.

Answer: B

Solution:

B-DNA: ~10.4 bp per turn.

Subject: Genetics | Topic: Genetics | Difficulty: difficult

Q30.

Answer: C

Solution:

$y = A\cos(2x) + B\sin(2x)$. $A=1$, $B=0$. $y(\pi/2) = \cos(\pi) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q31.

Answer: A

Solution:

$|\langle u, v \rangle| \leq \|u\| * \|v\|$. Equality when u and v are linearly dependent.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q32.

Answer: B

Solution:

ABO: I^A , I^B , i (three alleles).

Subject: Genetics | Topic: Genetics | Difficulty: easy

Q33.

Answer: A

Solution:

$f(x)=x$ is odd on symmetric interval. Fourier series has only sine terms.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q34.

Answer: A

Solution:

Eigenvalues of A^2 are 1, 4, 9. Trace = sum = $1+4+9 = 14$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q35.

Answer: A

Solution:

$L\{e^{(-at)}\} = 1/(s+a)$. Here $a=3$: $1/(s+3)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q36.

Answer: B

Solution:

$(1+1/n)^{2n} = [(1+1/n)^n]^2 \rightarrow e^2$ as $n \rightarrow \infty$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q37.

Answer: C

Solution:

Linked genes on same chromosome: recombination $< 50\%$.

Subject: Genetics | Topic: Genetics | Difficulty: difficult

Q38.

Answer: B

Solution:

$|x|$ is continuous everywhere. Left derivative at 0 = -1, right = +1. Not differentiable at 0.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q39.

Answer: C

Solution:

Rolle's: f continuous on $[a,b]$, differentiable on (a,b) , $f(a)=f(b)$. Then $f'(c)=0$ for some c in (a,b) .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q40.

Answer: A

Solution:

Eigenvalues are 3 and 1. $[1,0]^T$ is eigenvector for eigenvalue 1. $A^{50}[1,0]^T = [1,0]^T$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q41.

Answer: A

Solution:

Area = πr^2 . Circumference = $2\pi r$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q42.

Answer: A

Solution:

$f(i) = i^2 + 3i + 2 = -1 + 3i$. $\text{Re}(f(i)) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q43.

Answer: B

Solution:

$dy/y = dx/x$. $\ln(y) = \ln(x) + C$. $y = C \cdot x$. Wait: $y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow \ln y = \ln x + C \Rightarrow y = C \cdot x$. That's option A. Hmm, let me check: $y \, dx - x \, dy = 0 \Rightarrow y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow y = C \cdot x$. Answer: A.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q44.

Answer: C

Solution:

Kirchhoff: for K_n , spanning trees = $n^{n-2} = 4^2 = 16$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q45.

Answer: A

Solution:

Row 2 = 2*Row 1, Row 3 = 3*Row 1. All rows linearly dependent. Rank = 1.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q46.

Answer: B

Solution:

$P(A|B) = P(A \cap B)/P(B) = 0.2/0.4 = 0.5$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q47.

Answer: A

Solution:

$L\{t^2 f(t)\} = -d/ds(F(s))$. $F(s) = 1/(s+2)$. $-d/ds[1/(s+2)] = 1/(s+2)^2$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q48.

Answer: A

Solution:

Homogeneous ODE. Substituting $y=x*v$ and $y(1)=0$ gives $y=0$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q49.

Answer: A

Solution:

10% law: ~10% energy transferred between trophic levels (Lindeman).

Subject: Ecology | Topic: Ecology | Difficulty: moderate

Q50.

Answer: A

Solution:

Area = πr^2 . Circumference = $2\pi r$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q51.

Answer: B

Solution:

Logistic: $dN/dt = rN(1-N/K)$. K is carrying capacity.

Subject: Ecology | Topic: Ecology | Difficulty: moderate

Q52.

Answer: A

Solution:

$L\{e^{-at}\} = 1/(s+a)$. Here $a=3$: $1/(s+3)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q53.

Answer: A

Solution:

Competitive: V_{max} unchanged, K_m increases (more substrate needed).

Subject: Biology | Topic: Biology | Difficulty: moderate

Q54.

Answer: C

Solution:

Linked genes on same chromosome: recombination < 50%.

Subject: Genetics | Topic: Genetics | Difficulty: moderate

Q55.

Answer: A

Solution:

$|\langle u, v \rangle| \leq \|u\| \cdot \|v\|$. Equality when u and v are linearly dependent.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q56.

Answer: B

Solution:

$|x|$ is continuous everywhere. Left derivative at 0 = -1, right = +1. Not differentiable at 0.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q57.

Answer: B

Solution:

Light reactions: ATP + NADPH for Calvin cycle. O₂ is byproduct.

Subject: Biology | Topic: Biology | Difficulty: difficult

Q58.

Answer: B

Solution:

Logistic: $\frac{dN}{dt} = rN(1-N/K)$. K is carrying capacity.

Subject: Ecology | Topic: Ecology | Difficulty: easy

Q59.

Answer: C

Solution:

Linked genes on same chromosome: recombination < 50%.

Subject: Genetics | Topic: Genetics | Difficulty: easy

Q60.

Answer: B

Solution:

Logistic: $\frac{dN}{dt} = rN(1-N/K)$. K is carrying capacity.

Subject: Ecology | Topic: Ecology | Difficulty: moderate

Q61.

Answer: B

Solution:

DNA replication: S phase in nucleus. Mitochondrial DNA replicates separately.

Subject: Biology | Topic: Biology | Difficulty: easy

Q62.

Answer: A

Solution:

10% law: ~10% energy transferred between trophic levels (Lindeman).

Subject: Ecology | Topic: Ecology | Difficulty: moderate

Q63.

Answer: A

Solution:

Area = πr^2 . Circumference = $2\pi r$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q64.

Answer: B

Solution:

ABO: I^A, I^B, i (three alleles).

Subject: Genetics | Topic: Genetics | Difficulty: moderate

Q65.

Answer: A

Solution:

Competitive: V_{max} unchanged, K_m increases (more substrate needed).

Subject: Biology | Topic: Biology | Difficulty: difficult

PAPER BLUEPRINT

Subject Distribution:

Engineering Mathematics: 31 questions, 52 marks
Ecology: 9 questions, 14 marks
Genetics: 9 questions, 14 marks
Biology: 9 questions, 13 marks
General Aptitude: 1 questions, 1 marks
Evolution: 1 questions, 1 marks
Environmental Science: 1 questions, 1 marks
Cell Biology: 1 questions, 1 marks
Plant Physiology: 1 questions, 1 marks
Zoology: 1 questions, 1 marks
Behavioral Ecology: 1 questions, 1 marks

Difficulty Distribution:

Easy: 21 | Moderate: 35 | Difficult: 9

Statistics:

Total Questions: 65
Total Marks: 100

This blueprint is generated from the actual question distribution in this paper.